



TRAINING PROGRAMME

Classical AI

Search, logic and probabilistic reasoning: the foundations of artificial intelligence

90 HOURS

FOR University students in computer science and engineering



Mohamed Aziz BEN HAHHA

mohamedazizbenhaha.netlify.app

Overview

This one-semester academic course covers the foundations of artificial intelligence: rational agents, search and problem solving, game playing, knowledge representation and logic, and reasoning under uncertainty and over time. It follows the structure of Russell and Norvig's *Artificial Intelligence: A Modern Approach* (4th edition). Each part combines lectures that derive the algorithms with an implementation mini-project in Python, so that students both understand and build each method. The course closes with a capstone project: an AI system combining search, logic or probabilistic reasoning, proposed, developed and presented by each team.

At a glance

#	PHASE	FOCUS	HOURS
01	Introduction & Intelligent Agents	What AI is, agents and environments, rationality	9 h
02	Search & Problem Solving	Uninformed and heuristic search, local search, evolutionary methods, CSPs	18 h
03	Game Playing	Minimax, alpha-beta pruning, evaluation functions, Monte Carlo tree search	6 h
04	Knowledge Representation & Logic	Propositional and first-order logic, inference, knowledge engineering	15 h
05	Probabilistic Reasoning	Probability, Bayesian networks, exact and approximate inference	15 h
06	Reasoning over Time	Markov processes, hidden Markov models, filtering, smoothing, Viterbi	9 h
07	Final Project	A capstone AI system, proposed, developed and presented	9 h
	Buffer	Catch-up, revision, extra lab time	9 h
	Total		90 h

Key facts

AUDIENCE	University students in computer science and engineering
PREREQUISITES	Python programming (course: Python Programming Fundamentals), or equivalent; discrete mathematics and basic probability
DURATION	90 h: 81 h of chapters + 9 h buffer, over one academic semester
FORMAT	Lectures that derive each algorithm, followed by implementation mini-projects in Python
LAB ENVIRONMENT	One laptop per student with Python 3, Jupyter and VS Code; course code and datasets in a shared repository; no account or cost
ASSESSMENT	Six mini-projects and a capstone project presented to the class; certificate of completion for students who complete both; grading follows the host institution's rules
OUTCOME	Students can formulate AI problems, implement and compare search, logic and probabilistic algorithms, and justify which method fits a problem

PHASE 01

9 H

Introduction & Intelligent Agents

Students see what AI is, where it came from and where it is used, including its ethical questions, then learn the agent framework used throughout the course: environments, rationality and agent architectures. Mini-project: a simple agent designed and simulated in a grid environment.

#	CHAPTER	CONTENT	FORMAT	HOURS
1.1	What AI Is	Definitions, history, applications, ethics	Lecture + Workshop	3 h
1.2	Intelligent Agents	Agents, environments, rationality, agent architectures	Lecture + Lab	3 h
1.3	Mini-Project: Agents	Design and simulate a simple agent	Project	3 h

PHASE 02

18 H

Search & Problem Solving

Students formulate problems as state spaces and solve them with uninformed and heuristic search, local search, genetic algorithms and particle swarms, and constraint satisfaction. Mini-project: a planning or optimisation problem solved and compared with at least two search methods.

#	CHAPTER	CONTENT	FORMAT	HOURS
2.1	Uninformed Search	Problem formulation, BFS, DFS, uniform-cost search, iterative deepening	Lecture + Lab	3 h
2.2	Heuristic Search	Greedy best-first, A*, admissibility and consistency	Lecture + Lab	3 h
2.3	Local Search	Hill climbing, simulated annealing, local beam search	Lecture + Lab	3 h
2.4	Evolutionary & Swarm Methods	Genetic algorithms, particle swarm optimisation	Lecture + Lab	3 h
2.5	Constraint Satisfaction	CSPs, backtracking, arc consistency	Lecture + Lab	3 h
2.6	Mini-Project: Search	Solve a planning or optimisation problem with search	Project	3 h

MATHS COVERED IN THIS PHASE

Graph theory (state spaces, paths) · Complexity (time and space of search algorithms) · Optimisation (objective functions, local optima, annealing schedules)

PHASE 03

6 H

Game Playing

Students search against an opponent with minimax and alpha-beta pruning, design evaluation functions, and use Monte Carlo tree search when the game tree is too large. Mini-project: a game-playing agent evaluated against baseline players.

#	CHAPTER	CONTENT	FORMAT	HOURS
3.1	Adversarial Search	Minimax, alpha-beta pruning, evaluation functions, Monte Carlo tree search	Lecture + Lab	3 h
3.2	Mini-Project: Game Agent	Build and evaluate a game-playing agent	Project	3 h

MATHS COVERED IN THIS PHASE

Game theory (zero-sum games, minimax value) · Probability (Monte Carlo estimates, UCB1 in MCTS)

PHASE 04

15 H

Knowledge Representation & Logic

Students represent knowledge in propositional and first-order logic and derive new facts with resolution, DPLL, forward and backward chaining and unification. Mini-project: a small knowledge base and reasoner that answers queries in a chosen domain.

#	CHAPTER	CONTENT	FORMAT	HOURS
4.1	Propositional Logic	Syntax, semantics, entailment	Lecture + Lab	3 h
4.2	Propositional Inference	Resolution, DPLL, forward and backward chaining	Lecture + Lab	3 h
4.3	First-Order Logic	Syntax, semantics, knowledge engineering	Lecture + Lab	3 h
4.4	First-Order Inference	Unification, chaining, resolution	Lecture + Lab	3 h
4.5	Mini-Project: Knowledge-Based Agent	Build a small knowledge base and reasoner	Project	3 h

MATHS COVERED IN THIS PHASE

Formal logic (syntax, semantics, entailment, soundness and completeness) · Satisfiability · Unification

PHASE 05

15 H

Probabilistic Reasoning

Students reason under uncertainty with probability and Bayes' rule, model dependencies with Bayesian networks, and answer queries with exact inference and with sampling methods. Mini-project: a Bayesian network for a real decision problem, queried and checked against data.

#	CHAPTER	CONTENT	FORMAT	HOURS
5.1	Uncertainty & Probability	Probability axioms, Bayes' rule, independence	Lecture + Lab	3 h
5.2	Bayesian Networks	Structure, conditional independence, semantics	Lecture + Lab	3 h
5.3	Exact Inference	Enumeration, variable elimination	Lecture + Lab	3 h
5.4	Approximate Inference	Sampling, likelihood weighting, MCMC	Lecture + Lab	3 h
5.5	Mini-Project: Bayesian Network	Model and query a real decision problem	Project	3 h

MATHS COVERED IN THIS PHASE

Probability (axioms, conditional probability, Bayes' rule, independence) · Sampling and Markov chain Monte Carlo

PHASE 06

9 H

Reasoning over Time

Students model processes that change over time with Markov chains and hidden Markov models, and infer hidden states with filtering, smoothing and the Viterbi algorithm. Mini-project: sequence labelling or tracking with a hidden Markov model.

#	CHAPTER	CONTENT	FORMAT	HOURS
6.1	Temporal Models & HMMs	Markov processes, hidden Markov models	Lecture + Lab	3 h
6.2	Inference over Time	Filtering, smoothing, Viterbi	Lecture + Lab	3 h
6.3	Mini-Project: HMM	Sequence labelling or tracking with an HMM	Project	3 h

MATHS COVERED IN THIS PHASE

Markov chains · Dynamic programming (forward-backward, Viterbi)

PHASE 07

9 H

Final Project

Teams of two or three propose and build an AI system that combines at least two parts of the course, reusing their mini-projects, then present it to the class.

#	CHAPTER	CONTENT	FORMAT	HOURS
7.1	Proposal & Design	Problem statement, choice of methods, design and evaluation plan, approved by the lecturer	Project	3 h
7.2	Development	Implementation and experiments	Project	3 h
7.3	Evaluation & Report	Results, comparison with a baseline, written report	Project	1.5 h
7.4	Presentation	Presentation to the class; questions and feedback	Project	1.5 h

FINAL PROJECT

Capstone AI System

An AI system that combines at least two of the course's methods (search, game playing, logic, probabilistic or temporal reasoning) to solve a problem chosen by the team, evaluated against a baseline and explained in a written report.

STACK Python 3, Jupyter, NumPy, aima-python, NetworkX, pgmpy, hmmlearn

DELIVERABLE Proposal, code repository, written report, presentation